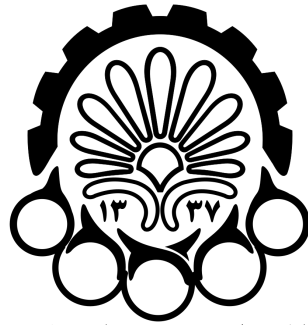




دانشگاه صنعتی امیرکبیر  
( پلی تکنیک تهران )

دانشکده مهندسی کامپیوتر

درس مبانی رایانش ابری

فاز اول پروژه

نگارنده:

علی ناصری نیا

۴۰۱۳۱۰۵۳

محمد امین فرح بخش

۴۰۱۳۱۰۲۹

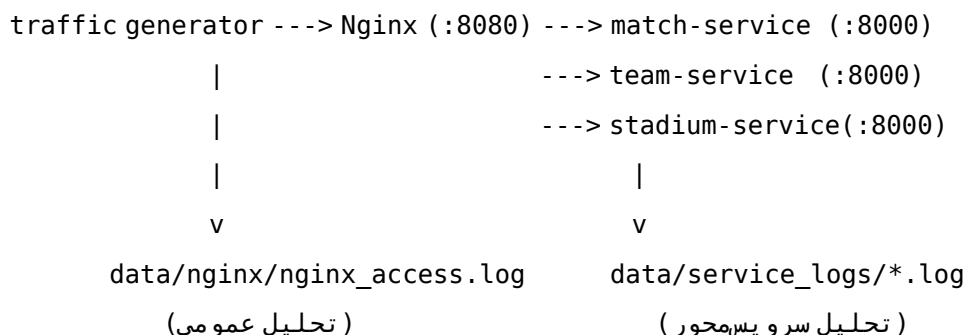
استاد راهنما:

دکتر جوادی

تیر ۱۴۰۵

## معماری

اجرای پروژه بر اساس معماری زیر طراحی شده است:



نکات کلیدی معماری:

- پورت سرویس‌های backend روی هاست publish نشده است، بنابراین تولیدکننده‌ی ترافیک به‌هیچ‌وجه نمی‌تواند Nginx را دور بزند.
- Nginx هدرهای x-request-id، x-client-country و x-scenario را به سرویس مقصد منتقل می‌کند و همان‌ها در لاگ ساختاریافته‌ی سرویس ثبت می‌شوند.
- Nginx در \$request\_time بر حسب ثانیه است و در Job 1 به request\_time\_ms تبدیل می‌شود.

## اجرای موفق سرویس‌ها

### سرویس match-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/matches?date=2026-06-25"
```

|         |            |             |           |                     |             |         |             |
|---------|------------|-------------|-----------|---------------------|-------------|---------|-------------|
| date    | 2026-06-25 |             |           |                     |             |         |             |
| count   | 3          |             |           |                     |             |         |             |
| matches | #          | team_a      | team_b    | stadium             | city        | country | stage       |
|         | 0          | Netherlands | Morocco   | BC Place Vancouver  | Vancouver   | Canada  | Group Stage |
|         | 1          | Belgium     | Australia | Kansas City Stadium | Kansas City | USA     | Group Stage |
|         | 2          | Croatia     | Italy     | Miami Stadium       | Miami       | USA     | Group Stage |

شکل ۱: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/matches?date=2026-06-"
Error: nu::shell::http_error

* HTTP Error 400 (Bad Request): http://localhost:8080/api/matches?date=2026-06-
[repl_entry #252:1:10]
1 http get "http://localhost:8080/api/matches?date=2026-06-"
.
.
.
{"error":"invalid date format, expected YYYY-MM-DD","received":"2026-06-"}

```

شکل ۲: تاریخ نامعتبر

## سرویس team-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/teams?name=Argentina"
```

|               |           |
|---------------|-----------|
| name          | Argentina |
| country_code  | ARG       |
| confederation | CONMEBOL  |

شکل ۳: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/teams?name=Argentin"
Error: nu::shell::http_error

* HTTP Error 404 (Not Found): http://localhost:8080/api/teams?name=Argentin
[repl_entry #268:1:10]
1 http get "http://localhost:8080/api/teams?name=Argentin"
└─ {"error": "team not found", "name": "Argentin"}
```

شکل ۴: کشور نامعتبر

## سرویس stadium-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> let params = { name: "New York New Jersey Stadium" }
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
```

|             |                             |
|-------------|-----------------------------|
| name        | New York New Jersey Stadium |
| common_name | MetLife Stadium             |
| city        | New York New Jersey         |
| country     | USA                         |
| capacity    | 82500                       |

شکل ۵: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> let params = { name: "ghale hasan khan" }
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
Error: nu::shell::http_error

* HTTP Error 404 (Not Found): http://localhost:8080/api/stadiums?name=ghale+hasan+khan
[repl_entry #300:1:10]
1 http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
└─ {"error": "stadium not found", "name": "ghale hasan khan"}
```

شکل ۶: ورزشگاه نامعتبر

## اجرای موفق کانتینرها

خروجی دستور docker compose ps نشان‌دهنده‌ی اجرای صحیح تمام کانتینرها است:

| # | NAME            | IMAGE                                               | COMMAND                  | SERVICE         | CREATED     | STATUS     | PORTS                                   |
|---|-----------------|-----------------------------------------------------|--------------------------|-----------------|-------------|------------|-----------------------------------------|
| 0 | match-service   | cc_project_phase1_40131053_40131029-match-service   | "sh -c 'uvicorn main..." | match-service   | 2 hours ago | Up 2 hours | 8000/tcp                                |
| 1 | nginx           | nginx:1.25-alpine                                   | "/docker-entrypoint..."  | nginx           | 2 hours ago | Up 2 hours | 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp |
| 2 | stadium-service | cc_project_phase1_40131053_40131029-stadium-service | "sh -c 'uvicorn main..." | stadium-service | 2 hours ago | Up 2 hours | 8000/tcp                                |
| 3 | team-service    | cc_project_phase1_40131053_40131029-team-service    | "sh -c 'uvicorn main..." | team-service    | 2 hours ago | Up 2 hours | 8000/tcp                                |

شکل ۷: وضعیت کانتینرها

همچنین آزمون صحت تنظیمات Nginx با موفقیت انجام شد:

docker compose exec nginx nginx -t

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose exec nginx nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

شکل ۸: آزمون تنظیمات Nginx

## تست درخواست از مسیر Nginx

جهت تست اولیه‌ی سیستم، یک درخواست نمونه به Nginx ارسال شد:

```
curl -H "X-Request-ID: manual-001" \
  -H "X-Client-Country: Iran" \
  -H "X-Scenario: normal" \
  "http://localhost:8080/api/teams?name=Argentina"
```

|               |           |
|---------------|-----------|
| name          | Argentina |
| country_code  | ARG       |
| confederation | CONMEBOL  |

شکل ۹: نتیجه‌ی درخواست تستی از مسیر nginx

## اجرای traffic generator

برای تولید بار سنگین بر روی سیستم، از اسکریپت تولیدکننده‌ی ترافیک استفاده شد:

```
python3 traffic-generator/generate.py --requests 10000 --nginx-url http://localhost:8080
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> python3 traffic-generator/generate.py --requests 10000 --nginx-url http://localhost:8080
Sending 10000 requests to http://localhost:8080 with 32 workers ...
 5000 / 10000 done (16s)
10000 / 10000 done (30s)

Finished 10000 requests in 30.2s (331 req/s)
 2xx/3xx : 8741
 4xx    : 888
 5xx    : 371
 failed : 0

Nginx gateway log : data/nginx/nginx_access.log
Service logs      : data/service_logs/*.log
```

شکل ۱۰: اجرای تولیدکننده‌ی ترافیک با ۱۰۰۰۰۰ درخواست

## بخشی از لاگ نهایی Nginx

پس از تولید ترافیک، لاگ Nginx با استفاده از دستور زیر مشاهده شد:

```
tail -n 3 data/nginx/nginx_access.log
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 3 data/nginx/nginx_access.log | lines | each { from json }
```

| # | timestamp                 | request_id | client_ip  | client_country | scenario | method | path                                  | service         | status_code | request_time_sec | user_agent        |
|---|---------------------------|------------|------------|----------------|----------|--------|---------------------------------------|-----------------|-------------|------------------|-------------------|
| 0 | 2026-07-09T21:34:50+00:00 | req_009908 | 172.18.0.1 | Japan          | slow     | GET    | /api/stadiums?name=Monterrey+Stadium  | stadium-service | 200         | 1.363            | traffic-generator |
| 1 | 2026-07-09T21:34:51+00:00 | req_009977 | 172.18.0.1 | Brazil         | slow     | GET    | /api/stadiums?name=Atlanta+Stadium    | stadium-service | 200         | 1.267            | traffic-generator |
| 2 | 2026-07-09T21:34:51+00:00 | req_009904 | 172.18.0.1 | USA            | slow     | GET    | /api/stadiums?name=BC+Place+Vancouver | stadium-service | 200         | 1.459            | traffic-generator |

شکل ۱۱: سه خط آخر لاگ Nginx

## بخشی از لاگ نهایی سرویس‌ها

به‌طور مشابه، لاگ سرویس‌ها نیز مشاهده شد:

```
tail -n 2 data/service_logs/team_service.log
```

```
tail -n 2 data/service_logs/match_service.log
```

```
tail -n 2 data/service_logs/stadium_service.log
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/team_service.log | lines | each { from json }
```

| # | timestamp                | request_id | client_country | scenario | service      | endpoint   | entity_type | entity_value | status_code | processing_time_ms | event_type  |
|---|--------------------------|------------|----------------|----------|--------------|------------|-------------|--------------|-------------|--------------------|-------------|
| 0 | 2026-07-09T21:34:50.010Z | req_009945 | Iran           | slow     | team-service | /api/teams | team        | Serbia       | 200         | 339                | team_lookup |
| 1 | 2026-07-09T21:34:50.492Z | req_009854 | Canada         | slow     | team-service | /api/teams | team        | Saudi Arabia | 200         | 1058               | team_lookup |

شکل ۱۲: دو خط آخر لاگ team\_service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/match_service.log | lines | each { from json }
```

| # | timestamp                | request_id | client_country | scenario | service       | endpoint     | entity_type | entity_value | status_code | processing_time_ms | event_type   |
|---|--------------------------|------------|----------------|----------|---------------|--------------|-------------|--------------|-------------|--------------------|--------------|
| 0 | 2026-07-09T21:34:50.391Z | req_009714 | USA            | slow     | match-service | /api/matches | match_day   | 2026-06-13   | 200         | 1297               | match_lookup |
| 1 | 2026-07-09T21:34:50.574Z | req_009876 | Brazil         | slow     | match-service | /api/matches | match_day   | 2026-06-22   | 200         | 1092               | match_lookup |

شکل ۱۳: دو خط آخر لاگ match-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/stadium_service.log | lines | each { from json }
```

| # | timestamp                | request_id | client_country | scenario | service         | endpoint      | entity_type | entity_value       | status_code | processing_time_ms | event_type     |
|---|--------------------------|------------|----------------|----------|-----------------|---------------|-------------|--------------------|-------------|--------------------|----------------|
| 0 | 2026-07-09T21:34:51.022Z | req_009977 | Brazil         | slow     | stadium-service | /api/stadiums | stadium     | Atlanta Stadium    | 200         | 1264               | stadium_lookup |
| 1 | 2026-07-09T21:34:51.025Z | req_009904 | USA            | slow     | stadium-service | /api/stadiums | stadium     | BC Place Vancouver | 200         | 1457               | stadium_lookup |

شکل ۱۴: دو خط آخر لاگ stadium\_service

## اجرای Hadoop Streaming

برای پردازش داده‌ها، Hadoop Streaming به‌کار گرفته شد:

```
docker compose -f hadoop/docker-compose.yml up -d
```

```
docker exec -it namenode bash /project/scripts/run_mapreduce.sh
```

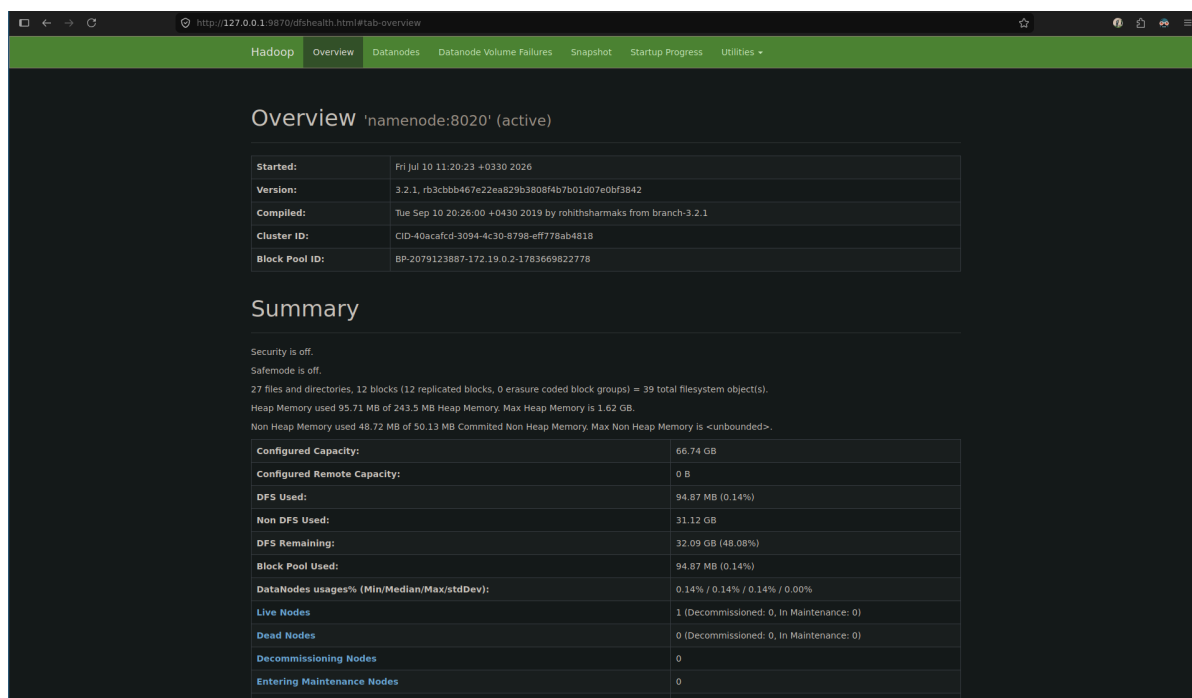
```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose -f hadoop/docker-compose.yml up -d
[+] up 21/21
✓ Image registry.docker.ir/bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Pulled
✓ Image registry.docker.ir/bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Pulled
✓ Network hadoop_default Created
✓ Container namenode Started
✓ Container datanode Started
```

شکل ۱۵: ساخته شدن Hadoop

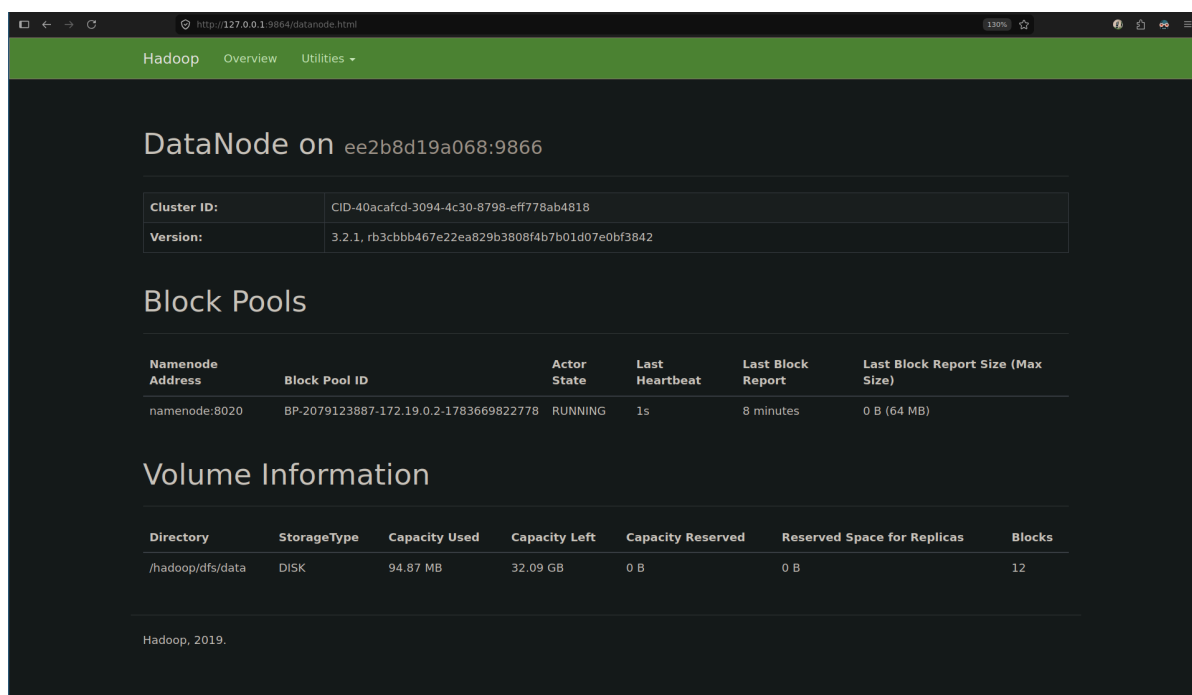
```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose -f hadoop/docker-compose.yml ps | detect columns --guess
```

| # | NAME     | IMAGE                                                              | COMMAND                | SERVICE  | CREATED       | STATUS                 | PORTS                                       |
|---|----------|--------------------------------------------------------------------|------------------------|----------|---------------|------------------------|---------------------------------------------|
| 0 | namenode | registry.docker.ir/bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 | "/entrypoint.sh sh -c" | namenode | 3 minutes ago | Up 3 minutes (healthy) | 0.0.0.0:9864->9864/tcp, [::]:9864->9864/tcp |
| 1 | namenode | registry.docker.ir/bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 | "/entrypoint.sh sh -c" | namenode | 3 minutes ago | Up 3 minutes (healthy) | 0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp |

شکل ۱۶: وضعیت کانترینرهای Hadoop



شکل ۱۷: رابط کاربری NameNode



شکل ۱۸: اجرای Job های Hadoop Streaming

`hdfs dfs -ls /input`

`hdfs dfs -ls /output`

```

root@a57ea740ff0c:/# hdfs dfs -ls /input
Found 2 items
-rw-r--r--  3 root supergroup  28882062 2026-07-10 07:50 /input/nginx_access.log
drwxr-xr-x  - root supergroup      0 2026-07-10 07:50 /input/service_logs
root@a57ea740ff0c:/#
root@a57ea740ff0c:/# hdfs dfs -ls /output
Found 5 items
drwxr-xr-x  - root supergroup      0 2026-07-10 07:50 /output/job1
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job2
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job3
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job4
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job5
root@a57ea740ff0c:/#

```

شکل ۱۹: محتویات HDFS قبل و بعد از پردازش

## خروجی‌های میانی

پس از پردازش، فایل‌های زیر به‌عنوان خروجی‌های میانی تولید شدند:

| توضیح                                                        | فایل                                  | Job |
|--------------------------------------------------------------|---------------------------------------|-----|
| لاگ دروازه‌ی تمیزشده، با request_time_ms                     | outputs/job1/cleanednginx_logs.csv    | 1   |
| لاگ سرویس‌ها با entity_type/entity_value                     | outputs/job1/cleaned_service_logs.csv | 1   |
| رکوردهای نامعتبر                                             | outputs/job1/invalid_logs.csv         | 1   |
| تعداد، خطای error rate، 4xx/5xx و میانگین زمان پاسخ هر سرویس | outputs/job2/service_stats.csv        | 2   |
| همان آمار به تفکیک endpoint                                  | outputs/job2/endpoint_stats.csv       | 2   |
| همان آمار به تفکیک سناریو                                    | outputs/job2/scenario_stats.csv       | 2   |
| تعداد درخواست هر کشور برای هر موجودیت                        | outputs/job3/country*requests.csv     | 3   |
| محبوب‌ترین موجودیت هر کشور                                   | outputs/job4/popular*by_country.csv   | 4   |

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> tree
.
├── final
│   └── summary.json
├── job1
│   ├── cleaned_nginx_logs.csv
│   ├── cleaned_service_logs.csv
│   └── invalid_logs.csv
├── job2
│   ├── endpoint_stats.csv
│   ├── scenario_stats.csv
│   └── service_stats.csv
├── job3
│   ├── country_city_requests.csv
│   ├── country_matchday_requests.csv
│   ├── country_stadium_requests.csv
│   └── country_team_requests.csv
└── job4
    ├── popular_city_by_country.csv
    ├── popular_matchday_by_country.csv
    ├── popular_stadium_by_country.csv
    └── popular_team_by_country.csv

5 directories, 15 files
```

شکل ۲۰: فایل‌های خروجی در پوشه‌ی outputs

نمونه‌ی outputs/job2/service\_stats.csv:

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> cat outputs/job2/service_stats.csv | from csv
```

| # | service         | total_requests | success_count | error_4xx | error_5xx | error_rate | avg_response_time_ms |
|---|-----------------|----------------|---------------|-----------|-----------|------------|----------------------|
| 0 | match-service   | 24564          | 21965         | 1592      | 1007      | 0.11       | 69.76                |
| 1 | stadium-service | 26378          | 21276         | 4261      | 841       | 0.19       | 165.28               |
| 2 | team-service    | 36008          | 32563         | 1913      | 1532      | 0.10       | 47.02                |

شکل ۲۱: outputs/job2/service\_stats.csv

نمونه‌ی outputs/job4/popular\_team\_by\_country.csv:

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> cat job4/popular_team_by_country.csv | from csv
```

| #  | country   | popular_team | total_requests |
|----|-----------|--------------|----------------|
| 0  | Argentina | Argentina    | 2488           |
| 1  | Brazil    | Brazil       | 2235           |
| 2  | Canada    | Canada       | 542            |
| 3  | France    | France       | 1572           |
| 4  | Germany   | Germany      | 2485           |
| 5  | Iran      | Argentina    | 3993           |
| 6  | Japan     | Japan        | 1226           |
| 7  | Mexico    | Mexico       | 1067           |
| 8  | Morocco   | Morocco      | 335            |
| 9  | Spain     | Spain        | 853            |
| 10 | USA       | USA          | 2180           |

شکل ۲۲: outputs/job4/popular\_team\_by\_country.csv



## خروجی نهایی — outputs/final/summary.json

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> cat final/summary.json | from json
```

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|----------------------|--------|----------------------|--------|----------------|-------|---------------|---------|-----------|-------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|-----------|-------|----------------------|---------|----------------|-------|---------------|-------|-----------|------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|-----------|------|----------------------|-------|----------------|-------|---------------|-------|-----------|------|
| <b>service_stats</b>                    | <table> <tr> <td><b>match-service</b></td><td> <table> <tr><td>error_rate</td><td>0.11</td></tr> <tr><td>error_5xx</td><td>1007</td></tr> <tr><td>avg_response_time_ms</td><td>69.76</td></tr> <tr><td>total_requests</td><td>24564</td></tr> <tr><td>success_count</td><td>21965</td></tr> <tr><td>error_4xx</td><td>1592</td></tr> </table> </td></tr> <tr> <td><b>stadium-service</b></td><td> <table> <tr><td>error_rate</td><td>0.19</td></tr> <tr><td>error_5xx</td><td>841</td></tr> <tr><td>avg_response_time_ms</td><td>165.28</td></tr> <tr><td>total_requests</td><td>26378</td></tr> <tr><td>success_count</td><td>21276</td></tr> <tr><td>error_4xx</td><td>4261</td></tr> </table> </td></tr> <tr> <td><b>team-service</b></td><td> <table> <tr><td>error_rate</td><td>0.10</td></tr> <tr><td>error_5xx</td><td>1532</td></tr> <tr><td>avg_response_time_ms</td><td>47.02</td></tr> <tr><td>total_requests</td><td>36008</td></tr> <tr><td>success_count</td><td>32563</td></tr> <tr><td>error_4xx</td><td>1913</td></tr> </table> </td></tr> </table> | <b>match-service</b> | <table> <tr><td>error_rate</td><td>0.11</td></tr> <tr><td>error_5xx</td><td>1007</td></tr> <tr><td>avg_response_time_ms</td><td>69.76</td></tr> <tr><td>total_requests</td><td>24564</td></tr> <tr><td>success_count</td><td>21965</td></tr> <tr><td>error_4xx</td><td>1592</td></tr> </table> | error_rate | 0.11      | error_5xx            | 1007   | avg_response_time_ms | 69.76  | total_requests | 24564 | success_count | 21965   | error_4xx | 1592  | <b>stadium-service</b> | <table> <tr><td>error_rate</td><td>0.19</td></tr> <tr><td>error_5xx</td><td>841</td></tr> <tr><td>avg_response_time_ms</td><td>165.28</td></tr> <tr><td>total_requests</td><td>26378</td></tr> <tr><td>success_count</td><td>21276</td></tr> <tr><td>error_4xx</td><td>4261</td></tr> </table> | error_rate | 0.19   | error_5xx | 841   | avg_response_time_ms | 165.28  | total_requests | 26378 | success_count | 21276 | error_4xx | 4261 | <b>team-service</b> | <table> <tr><td>error_rate</td><td>0.10</td></tr> <tr><td>error_5xx</td><td>1532</td></tr> <tr><td>avg_response_time_ms</td><td>47.02</td></tr> <tr><td>total_requests</td><td>36008</td></tr> <tr><td>success_count</td><td>32563</td></tr> <tr><td>error_4xx</td><td>1913</td></tr> </table> | error_rate | 0.10 | error_5xx | 1532 | avg_response_time_ms | 47.02 | total_requests | 36008 | success_count | 32563 | error_4xx | 1913 |
| <b>match-service</b>                    | <table> <tr><td>error_rate</td><td>0.11</td></tr> <tr><td>error_5xx</td><td>1007</td></tr> <tr><td>avg_response_time_ms</td><td>69.76</td></tr> <tr><td>total_requests</td><td>24564</td></tr> <tr><td>success_count</td><td>21965</td></tr> <tr><td>error_4xx</td><td>1592</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | error_rate           | 0.11                                                                                                                                                                                                                                                                                           | error_5xx  | 1007      | avg_response_time_ms | 69.76  | total_requests       | 24564  | success_count  | 21965 | error_4xx     | 1592    |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_rate                              | 0.11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_5xx                               | 1007                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| avg_response_time_ms                    | 69.76                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| total_requests                          | 24564                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| success_count                           | 21965                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_4xx                               | 1592                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>stadium-service</b>                  | <table> <tr><td>error_rate</td><td>0.19</td></tr> <tr><td>error_5xx</td><td>841</td></tr> <tr><td>avg_response_time_ms</td><td>165.28</td></tr> <tr><td>total_requests</td><td>26378</td></tr> <tr><td>success_count</td><td>21276</td></tr> <tr><td>error_4xx</td><td>4261</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | error_rate           | 0.19                                                                                                                                                                                                                                                                                           | error_5xx  | 841       | avg_response_time_ms | 165.28 | total_requests       | 26378  | success_count  | 21276 | error_4xx     | 4261    |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_rate                              | 0.19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_5xx                               | 841                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| avg_response_time_ms                    | 165.28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| total_requests                          | 26378                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| success_count                           | 21276                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_4xx                               | 4261                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>team-service</b>                     | <table> <tr><td>error_rate</td><td>0.10</td></tr> <tr><td>error_5xx</td><td>1532</td></tr> <tr><td>avg_response_time_ms</td><td>47.02</td></tr> <tr><td>total_requests</td><td>36008</td></tr> <tr><td>success_count</td><td>32563</td></tr> <tr><td>error_4xx</td><td>1913</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | error_rate           | 0.10                                                                                                                                                                                                                                                                                           | error_5xx  | 1532      | avg_response_time_ms | 47.02  | total_requests       | 36008  | success_count  | 32563 | error_4xx     | 1913    |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_rate                              | 0.10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_5xx                               | 1532                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| avg_response_time_ms                    | 47.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| total_requests                          | 36008                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| success_count                           | 32563                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| error_4xx                               | 1913                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>total_4xx</b>                        | 7766                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>most_popular_team_overall</b>        | Argentina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>predicted_final_winner</b>           | Argentina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>predicted_final</b>                  | France vs Argentina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>most_requested_match_day_overall</b> | 2026-06-25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| <b>popular_team_by_country</b>          | <table> <tr><td>Iran</td><td>Argentina</td></tr> <tr><td>Argentina</td><td>Argentina</td></tr> <tr><td>Mexico</td><td>Mexico</td></tr> <tr><td>Brazil</td><td>Brazil</td></tr> <tr><td>USA</td><td>USA</td></tr> <tr><td>Germany</td><td>Germany</td></tr> <tr><td>Spain</td><td>Spain</td></tr> <tr><td>Canada</td><td>Canada</td></tr> <tr><td>France</td><td>France</td></tr> <tr><td>Japan</td><td>Japan</td></tr> <tr><td>Morocco</td><td>Morocco</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Iran                 | Argentina                                                                                                                                                                                                                                                                                      | Argentina  | Argentina | Mexico               | Mexico | Brazil               | Brazil | USA            | USA   | Germany       | Germany | Spain     | Spain | Canada                 | Canada                                                                                                                                                                                                                                                                                         | France     | France | Japan     | Japan | Morocco              | Morocco |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Iran                                    | Argentina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Argentina                               | Argentina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Mexico                                  | Mexico                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Brazil                                  | Brazil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| USA                                     | USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Germany                                 | Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Spain                                   | Spain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Canada                                  | Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| France                                  | France                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Japan                                   | Japan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |
| Morocco                                 | Morocco                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |                                                                                                                                                                                                                                                                                                |            |           |                      |        |                      |        |                |       |               |         |           |       |                        |                                                                                                                                                                                                                                                                                                |            |        |           |       |                      |         |                |       |               |       |           |      |                     |                                                                                                                                                                                                                                                                                                |            |      |           |      |                      |       |                |       |               |       |           |      |

شکل ۲۳: محتوای فایل summary.json

تفسیر:

- **most\_requested\_service**: پرمراجعه‌ترین سرویس (از لاگ Nginx).
- **highest\_error\_rate\_service**: سرویسی که بیشترین نسبت خطا دارد؛ چون ترافیک `invalid_input` بیشتر به سمت `stadium-service` هدایت شده، این سرویس بالاترین `error rate` را دارد.
- **slowest\_endpoint: endpoint**: با بیشترین میانگین زمان پاسخ.
- **most\_popular\_team\_overall / ...\_match\_day... / ...\_stadium...** — از جمع خروجی Job 3 روی همه‌ی کشورها.
- **popular\_team\_by\_country**: خروجی Job 4.
- **predicted\_\***: از داده‌ی پیش‌بینی مسابقات (`data/predictions/`).

## بخش امتیازی — Spark Structured Streaming

برای پردازش زمان واقعی از Spark Streaming استفاده شد:

```
docker compose -f spark/docker-compose.yml run --rm spark /opt/spark/bin/spark-submit --master 'local[*]' /project/spark/streaming_app.py --input /project/data/stream/nginx --service-input /project/data/stream/service_logs --checkpoint /project/checkpoints/spark
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029:
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029:
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029: docker compose -f spark/docker-compose.yml run --rm spark /opt/spark/bin/spark-submit --master 'local[*]' /project/spark/streaming_app.py --input /project/data/stream/nginx --service-input /project/data/stream/service_logs --checkpoint /project/checkpoints/spark
Container spark-spark-run-4d37cca2d5d6 Created
26/07/10 13:24:05 INFO SparkContext: Running Spark version 3.5.1
26/07/10 13:24:05 INFO SparkContext: OS info Linux, 6.8.0-110-generic, amd64
26/07/10 13:24:05 INFO SparkContext: Java version 11.0.22
26/07/10 13:24:05 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/07/10 13:24:05 INFO ResourceUtils: =====
26/07/10 13:24:05 INFO ResourceUtils: No custom resources configured for spark.driver.
26/07/10 13:24:05 INFO ResourceUtils: =====
26/07/10 13:24:05 INFO SparkContext: Submitted application: worldcup-live-log-analytics
26/07/10 13:24:05 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , off
ap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/07/10 13:24:05 INFO ResourceProfile: Limiting resource is cpu
26/07/10 13:24:05 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/07/10 13:24:05 INFO SecurityManager: Changing view acls to: root
26/07/10 13:24:05 INFO SecurityManager: Changing modify acls to: root
26/07/10 13:24:05 INFO SecurityManager: Changing view acls groups to:
26/07/10 13:24:05 INFO SecurityManager: Changing modify acls groups to:
26/07/10 13:24:05 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; groups with view permissions: EMPTY; users with modify permissions: root; gr
ps with modify permissions: EMPTY
26/07/10 13:24:05 INFO Utils: Successfully started service 'sparkDriver' on port 33051.
26/07/10 13:24:05 INFO SparkEnv: Registering MapOutputTracker
26/07/10 13:24:05 INFO SparkEnv: Registering BlockManagerMaster
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/07/10 13:24:05 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/07/10 13:24:05 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-0eadf09e-2138-48db-ae1a-16e02dde8ce0
26/07/10 13:24:05 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/07/10 13:24:05 INFO SparkEnv: Registering OutputCommitCoordinator
26/07/10 13:24:05 INFO JettyUtil: Start Jetty 9.4.0.v20180605 for SparkUI
26/07/10 13:24:05 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/07/10 13:24:05 INFO Executor: Starting executor ID driver on host da82b41569de
26/07/10 13:24:05 INFO Executor: OS info Linux, 6.8.0-110-generic, amd64
26/07/10 13:24:05 INFO Executor: Java version 11.0.22
26/07/10 13:24:05 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/07/10 13:24:05 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@4be31dca for default.
26/07/10 13:24:05 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 40307.
26/07/10 13:24:05 INFO NettyBlockTransferService: Server created on da82b41569de:40307
26/07/10 13:24:05 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/07/10 13:24:05 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: Registering BlockManager da82b41569de:40307 with 424.4 MiB RAM, BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
Streaming queries started: windowed service stats, live error rate, busiest endpoint, popular_team_by_country_live
Drop new *.jsonl files into /project/data/stream/nginx to see the numbers update.
```

شکل ۲۴: ساخت docker compose

| Busiest service / endpoint (micro-batch 16) |                     |                 |          |                      |                |
|---------------------------------------------|---------------------|-----------------|----------|----------------------|----------------|
| window_start                                | window_end          | service         | requests | avg_response_time_ms | error_rate_pct |
| 2026-07-09 21:30:10                         | 2026-07-09 21:30:20 | team-service    | 335      | 69.4                 | 8.96           |
| 2026-07-09 21:30:10                         | 2026-07-09 21:30:20 | match-service   | 223      | 71.1                 | 9.42           |
| 2026-07-09 21:30:00                         | 2026-07-09 21:30:10 | team-service    | 1432     | 42.6                 | 9.5            |
| 2026-07-09 21:30:00                         | 2026-07-09 21:30:10 | match-service   | 983      | 63.9                 | 11.09          |
| 2026-07-09 21:30:10                         | 2026-07-09 21:30:20 | stadium-service | 236      | 166.9                | 19.07          |
| 2026-07-09 21:30:00                         | 2026-07-09 21:30:10 | stadium-service | 1036     | 179.9                | 19.02          |

| service         | endpoint      | requests |
|-----------------|---------------|----------|
| team-service    | /api/teams    | 2815     |
| stadium-service | /api/stadiums | 2032     |
| match-service   | /api/matches  | 1952     |
| nginx           | /health       | 1        |

شکل ۲۵: اجرای برنامه‌ی Spark Streaming

سورس کد پروژه

<https://github.com/alinaserinia6/Cloud-Computing-Project-1/tree/main>